PROJECT OVERVIEW DOCUMENT · HARDWARE DESIGN PROPOSAL

SMPS-CTRL

Modular Multi-Output DC Power Supply with STM32 Control and Monitoring

Design and Development of an STM32-Controlled Modular DC Power Supply



Conceptual illustration – not a photograph of a real device

"A modular multi-output DC power platform with intelligent control and real-time monitoring."

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All images in this document are conceptual illustrations. The device has not been built, tested, calibrated or certified.

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How to read this document

This document is written so that a reader who has never heard of the project can still understand **what the device is, how it works, what it looks like and how it will go from idea to a real product**. The first part (sections 2–4) is for all readers; the middle part (sections 5–10) goes into technical depth; the last part (sections 11–18) covers mechanical construction, roadmap, validation and the final product.

Label	Meaning
Confirmed requirement	Core requirement set by the project owner, used as the design basis.
Proposed target	Parameter or solution proposed to meet the requirements; may be adjusted after deeper study.
Needs validation	Value or capability not yet demonstrated; considered achieved only after experimental measurement.
Principle illustration / Concept model	A drawing that explains a concept. Not a photograph, manufacturing drawing or measured data.

Conventions: a period (.) is used as the decimal separator (e.g. 3.3V); international technical terms are kept in parentheses where they first appear. A glossary is provided in the appendix at the end of the document.

PART 02

Executive Summary

Executive Summary

Electronics work, especially embedded-systems development, always starts with a deceptively simple question: how do I power the circuit? An STM32 board needs 3.3V, sensors and USB parts need 5V, relays or fans need 12V, and the circuit under test needs an adjustable supply with a current limit so it does not "burn" when wired wrongly. On the lab bench this usually ends in a pile of adapters, cheap step-down modules and tangled wires.

SMPS-CTRL is a design proposal for a desktop DC power supply that gathers these needs into one device: four independent outputs (12V, 5V, 3.3V and one adjustable 1.8–12V channel with current limiting), fed from a commercial isolated 24V AC-DC supply, and **monitored, configured and fault-reported by an STM32 microcontroller**.

What the device should do

- Power several parts of a test bench at once: 3.3V/3A, 5V/3A, 12V/1A and one adjustable 1.8–12V/1A channel.
- The adjustable channel works in constant-voltage (CV) or constant-current (CC) mode to protect the circuit under test.
- Measure and display voltage, current and temperature of each channel; allow each channel to be switched on/off; report faults clearly.
- Communicate with a computer over USB for monitoring and configuration.
- Modular structure: each function sits on its own board, making repair, replacement and experimentation easy.

What gives the project its technical value

The project does not aim to assemble ready-made modules. The custom design effort focuses on **low-voltage DC-DC conversion, protection, measurement and embedded control** – the parts that demonstrate real engineering competence: designing a switching step-down (Buck) converter, high-frequency PCB layout, analog current/voltage sensing circuits, CV/CC control loops, thermal management, monitoring firmware and a validation process. The 230V AC mains part is handed to an enclosed commercial isolated supply, keeping the prototype safe and focused on the part with academic and engineering value.

Foundational design principle
Fast switching regulation and the critical protections are handled by **dedicated power-management ICs and analog circuits**. The STM32 acts as a **high-level supervisor** – it does not replace hardware protection.

4	48.9 W	24 V	5
independent output channels	maximum output power (sum of channels)	DC distribution bus	main hardware modules

Current status
This document describes a **design proposal**. No device has been built, fully simulated, measured or certified. All performance figures, protection thresholds and fault-isolation capabilities are targets that require experimental validation.

Target specifications

The table below lists the parameters from the project brief. The last column classifies each row as a confirmed requirement, a proposed target or an item needing validation; this classification is proposed by this document and the project owner may adjust it.

Parameter	Initial target	Classification
Nominal DC input voltage	24V DC	**Confirmed requirement**
Extended input voltage range (proposed)	18–30V DC, pending validation	**Needs validation**
Commercial AC-DC source	230V AC → 24V DC, isolated	**Confirmed requirement**
Proposed AC-DC power rating	100W	**Proposed target**
Channel 1	12V / 1A	**Confirmed requirement**
Channel 2	5V / 3A	**Confirmed requirement**
Channel 3	3.3V / 3A	**Confirmed requirement**
Channel 4	Adjustable 1.8–12V / 1A	**Proposed target**
Programmable output modes	Constant Voltage (CV) / Constant Current (CC)	**Proposed target**
Main controller	STM32	**Confirmed requirement**
Communication	USB; RS485 (optional)	**Proposed target**
Monitoring	Voltage, current, temperature	**Confirmed requirement**
Protection	Overcurrent, short circuit, overtemperature, overvoltage	**Confirmed requirement**
Construction	Modular PCB architecture	**Confirmed requirement**
Enclosure	Desktop enclosure, removable cover	**Confirmed requirement**

PART 03

Motivation and Practical Applications

Motivation and Practical Applications

A typical test bench often has many separate supplies. Each one solves one job but together they create a series of limitations in long-term use:

Common approach	Practical limitation
Many separate adapters (5V, 12V, 24V…)	Many sockets and cables; no central measurement; hard to tell which supply is overloaded or shorted.
Cheap step-down (DC-DC) modules	Unclear quality and ripple; often incomplete protection; switching noise easily disturbs sensitive circuits.
Conventional laboratory power supply	Adjustable but usually only 1–2 channels; no ready fixed channels for 3.3V/5V/12V; hard to integrate monitoring with a computer.
Powering from a PC USB port	Limited current and voltage; no independent protection for each device.

SMPS-CTRL proposes to solve these limitations with a device that has **several independent channels, individual protection, shared monitoring and a clear interface**.

Four typical use cases

Channel	Powers	Concrete examples
3.3V / 3A	STM32 development board and 3.3V logic circuits	Nucleo/Discovery boards, I²C sensors, ESP32 modules
5V / 3A	Sensors, USB hub, interface circuits	Distance sensors, small displays, USB-UART converters
12V / 1A	Actuators and peripherals	Relays, small fans, solenoid valves, low-power DC motors
ADJ 1.8–12V / 1A	Circuit under test, current-limited	Try a new circuit at several voltages; stop at the current limit instead of burning components

Illustrative scenario: a sensor test session

Suppose an STM32-based sensor node is being developed. The user plugs the STM32 board into the 3.3V channel, the sensor into the 5V channel and a cooling fan into the 12V channel. The new circuit under test – not yet known to be free of shorts – is connected to the adjustable channel with a 0.2A current limit. A computer connected over USB shows each channel's current; if the test circuit shorts, the ADJ channel switches to current limiting and the other channels continue to be monitored (design objective).

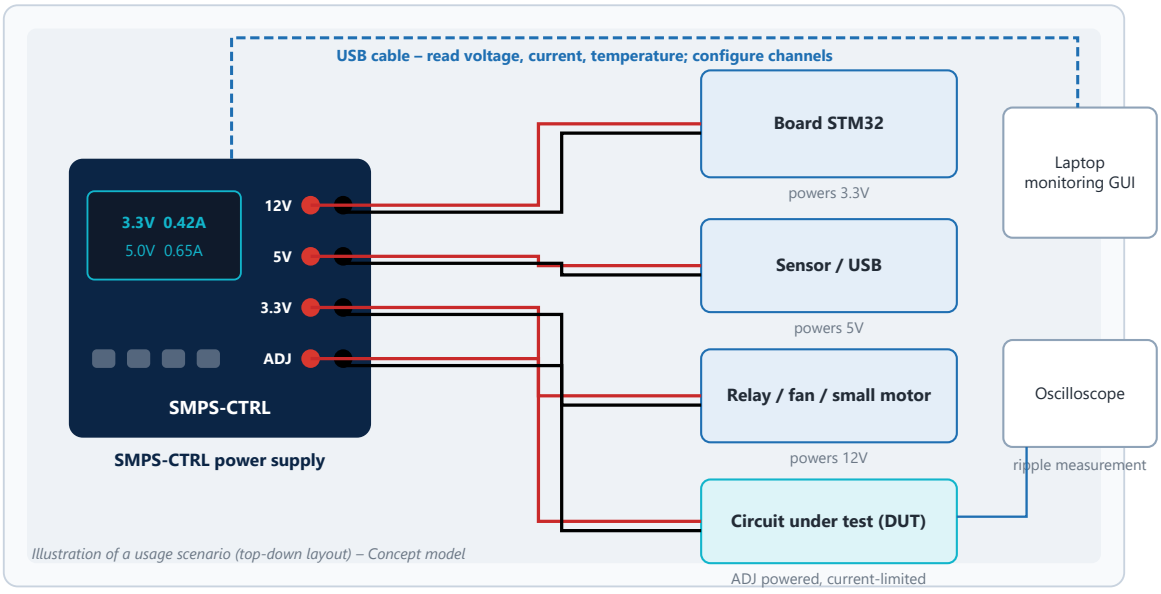


Figure 1. Illustration of a real laboratory usage scenario: four channels power four different groups of loads, and a computer monitors over USB. **[Concept model]**

PART 04

Overall System Architecture

Overall System Architecture

The block diagram below is the “map” of the whole project. Read it from top to bottom: mains power passes through the commercial AC-DC supply and becomes 24V DC; the 24V is protected and distributed to three fixed Buck converters, one CV/CC Buck converter and an independent auxiliary supply for the STM32. The STM32 measures, controls and reports faults over signal lines that are not on the power path.

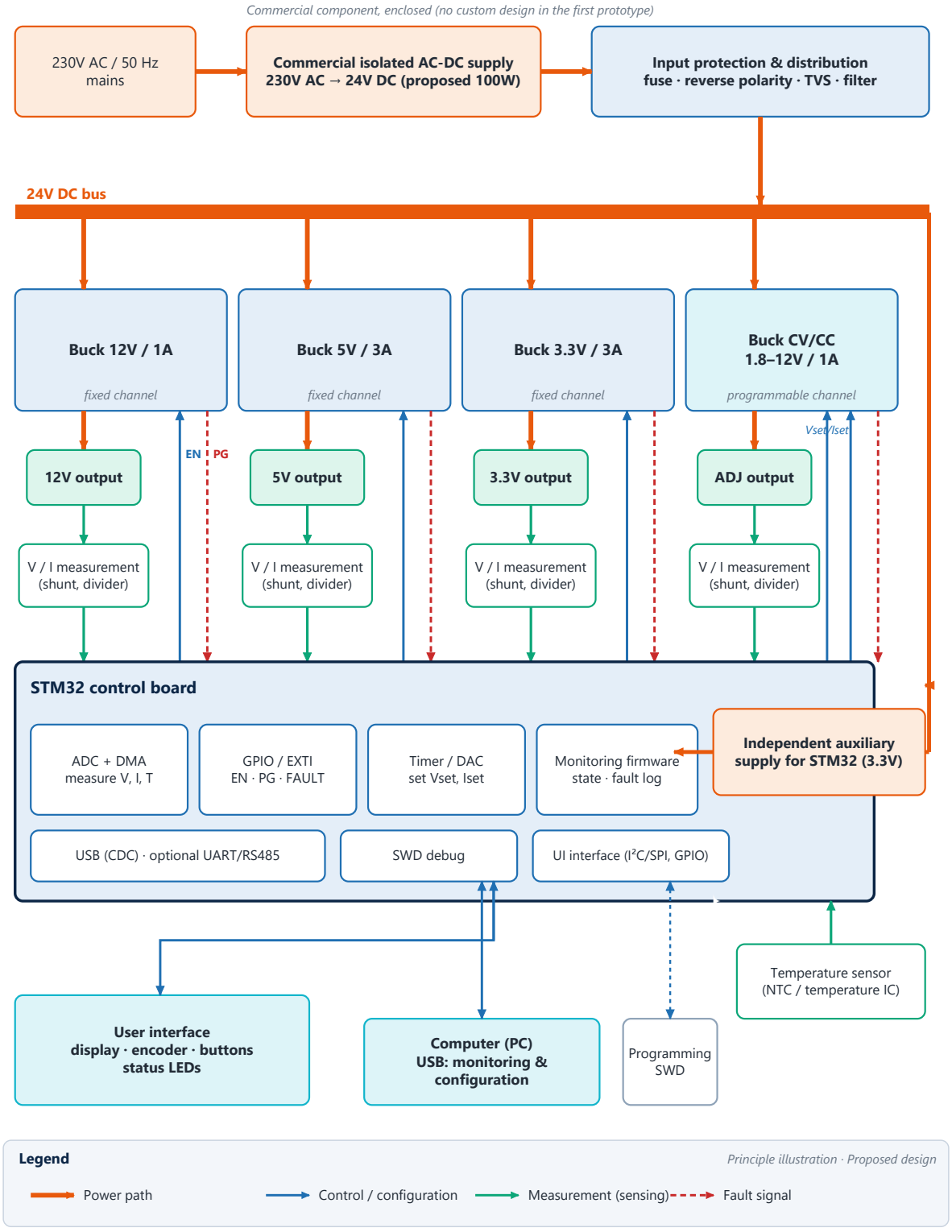


Figure 2. SMPS-CTRL system block diagram. Power paths (orange, thick) are separated from control paths (blue), measurement paths (green) and fault signals (red, dashed). **[Proposed design]**

PART 05

Electrical Power Architecture

Electrical Power Architecture

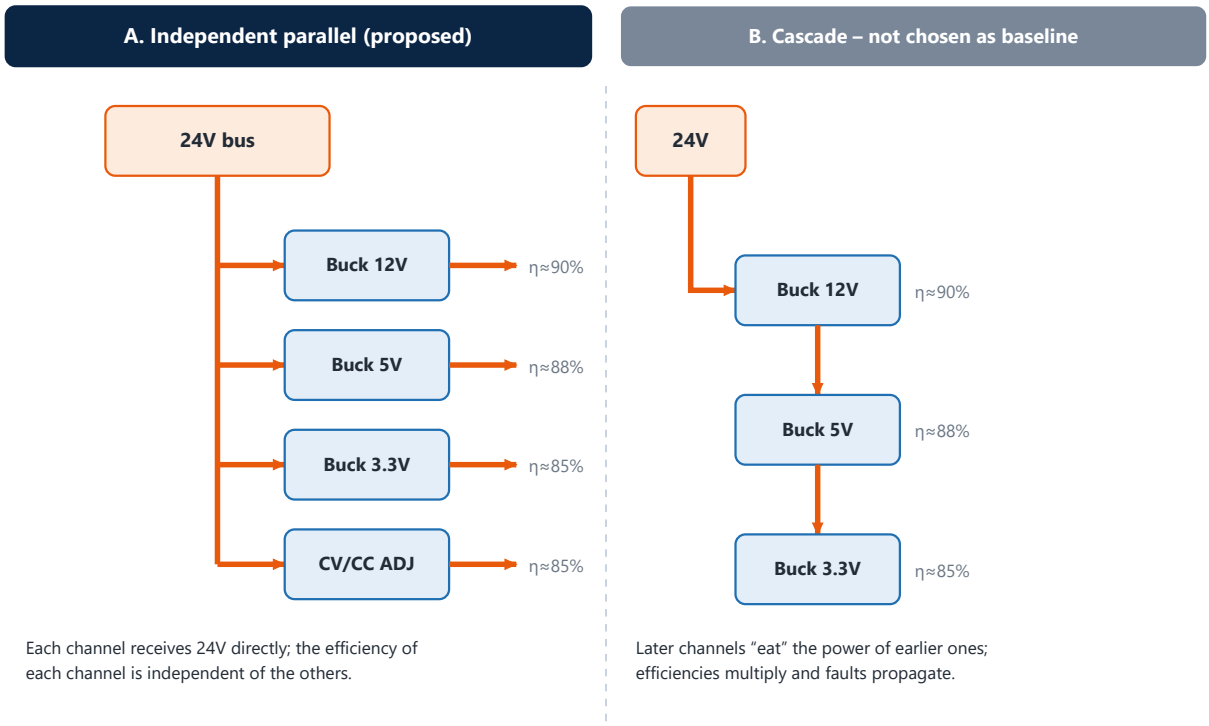
Why a 24V distribution bus

For the same power, the higher the voltage, the smaller the current (P = U × I). At 24V the total estimated input current is only about 2.5A (59.15W / 24V), whereas distributing the same power at 5V would require more than 11A. Lower current allows lighter wires, connectors and PCB traces, with less voltage drop and less heating. 24V is also a very common voltage for commercial AC-DC supplies, so it is easy to buy and replace.

Why independent parallel Buck converters

Instead of stepping down in series 24V → 12V → 5V → 3.3V, each channel receives **24V directly** and steps down once. This has four benefits:

- **Better efficiency:** each channel passes through only one conversion stage; efficiencies are not multiplied together (illustrative example: 0.90 × 0.88 × 0.85 ≈ 67% at the end of the chain, versus ≈ 85% for direct step-down).
- **Limited influence between channels:** a fault or heavy load on one channel is less likely to spread to others – a design objective that requires validation.
- **Easier debugging:** each channel can be tested, replaced and optimized separately.
- **Flexible expansion:** adding a new channel only requires adding a module connected to the 24V bus.



Criterion	Song song	Cascade
Efficiency at 3.3V (illustrative example)	≈ 85%	≈ 0.90×0.88×0.85 ≈ 67%
Effect when one channel fails	Limited to the channel (needs validation)	Affects the downstream channels
Debugging / replacing each module	Easy, independent	Chain-dependent
Number of conversion stages on the power path	1	Up to 3

Efficiencies in the figure are preliminary assumptions for comparison, not measured results.

Figure 3. Comparison of the independent parallel architecture (proposed) and the cascaded architecture. Efficiencies shown are illustrative assumptions, not measured results. **[Principle illustration]**

The cascaded architecture is not excluded in every case, but it is not suitable as the baseline for this project: it reduces efficiency, makes a fault in an earlier stage affect later stages and complicates the investigation of individual channels.

A separate supply for the STM32 and the benefits of modularity

Independent auxiliary supply for the STM32: if the microcontroller were powered from one of the power channels, a fault in that channel would also remove the "brain" exactly when it is needed to report the fault. A small dedicated auxiliary supply, taken from the 24V bus after the input protection stage, lets the STM32 keep running and record the event.

Modularity: each function is a separate board connected by connectors. When one board needs repair or improvement, only that board is replaced; to try a different Buck IC, only one board has to be redone instead of the whole system. This is also a familiar practice in industrial products.

PART 06

Output Channels and Power Budget

Output Channels and Power Budget

The four output channels serve four different groups of needs. Three fixed channels are simple and stable; the fourth is adjustable with current limiting.

Channel	Target specifications	Maximum power	Intended use
CH1	12V / 1A	12 × 1 = 12.0 W	Actuators, fans, relays
CH2	5V / 3A	5 × 3 = 15.0 W	Sensors, USB, interface circuits
CH3	3.3V / 3A	3.3 × 3 = 9.9 W	Bo STM32, logic 3.3V
CH4	Adjustable 1.8–12V / 1A	12 × 1 = 12.0 W (at 12V)	Circuit under test, current-limited
Total		**48.9 W**	Maximum output power if all four channels are fully loaded

Note
The power of CH4 at lower voltages is below 12W (for example 5V × 1A = 5W). The 48.9W figure is the upper limit when CH4 operates at 12V/1A.

From output power to input power

A Buck converter is not 100% efficient: part of the energy becomes heat. To know how many watts the 24V supply must provide, we calculate backwards from the output using **preliminary assumed efficiencies** (not measured results):

$$P_{in} = P_{out} / \eta$$

Channel	P ra (W)	Assumed η	P in calculation	P in (W)	Loss (W)
12V / 1A	12.00	90%	12.00 / 0.90	13.33	1.33
5V / 3A	15.00	88%	15.00 / 0.88	17.05	2.05
3.3V / 3A	9.90	85%	9.90 / 0.85	11.65	1.75
ADJ (12V / 1A)	12.00	85%	12.00 / 0.85	14.12	2.12
Auxiliary (STM32, interface...)	–	–	3 W allowance	3.00	3.00
Total	**48.90**		13.33 + 17.05 + 11.65 + 14.12 + 3.00	**≈ 59.15**	≈ 10.25

Summing the rounded values gives 59.15W; adding first and rounding afterwards gives 59.14W. This document uses ≈ 59.15W for consistency with the project brief.

25% design allowance: 59.15 W × 1.25 ≈ **73.94 W** – this is the preliminary target when selecting the AC-DC supply. Estimated input current at 24V: 59.15 / 24 ≈ 2.46A; at the target with allowance ≈ 3.08A.

Why a commercial 24V / 100W supply is proposed

- The 73.94W target does not match a standard power rating; 100W is the nearest commercial rating above it and is commonly available at 24V (4.17A).
- The extra capacity lets the supply operate at about 60% load, running cooler and lasting longer; it also allows for inrush current, load disturbances and real efficiency deviating from the assumptions.
- It allows later expansion (adding modules) without changing the supply.
- When selecting a specific model, conditions to check include: isolation, temperature range, protections declared by the manufacturer, size and how it is enclosed in the case – **no model has been selected** at this stage.

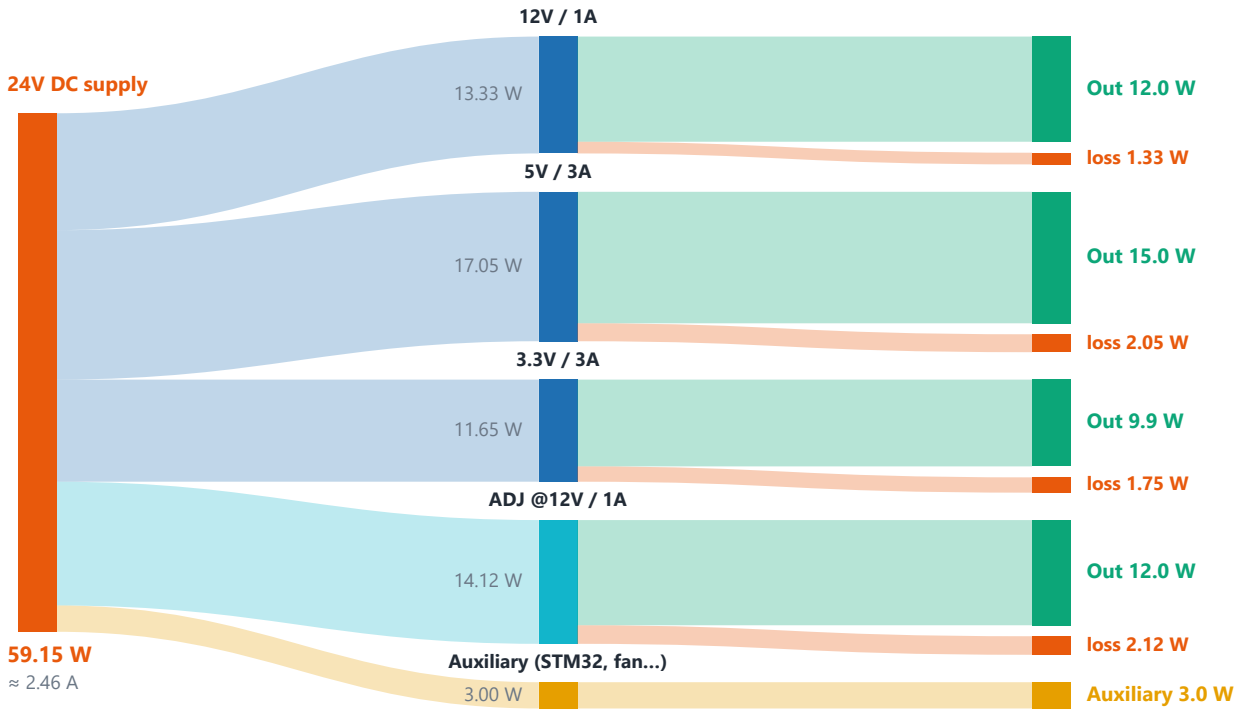


Figure 4. Estimated power flow from the 24V bus to the channels: green is useful power, orange is assumed thermal loss, yellow is auxiliary power. Preliminary estimate, not measured results. **[Preliminary estimate]**



Figure 5. Comparison of power levels: total maximum output power, estimated input power, the target with a 25% allowance and the proposed 100W supply. **[Preliminary estimate]**

Needs validation

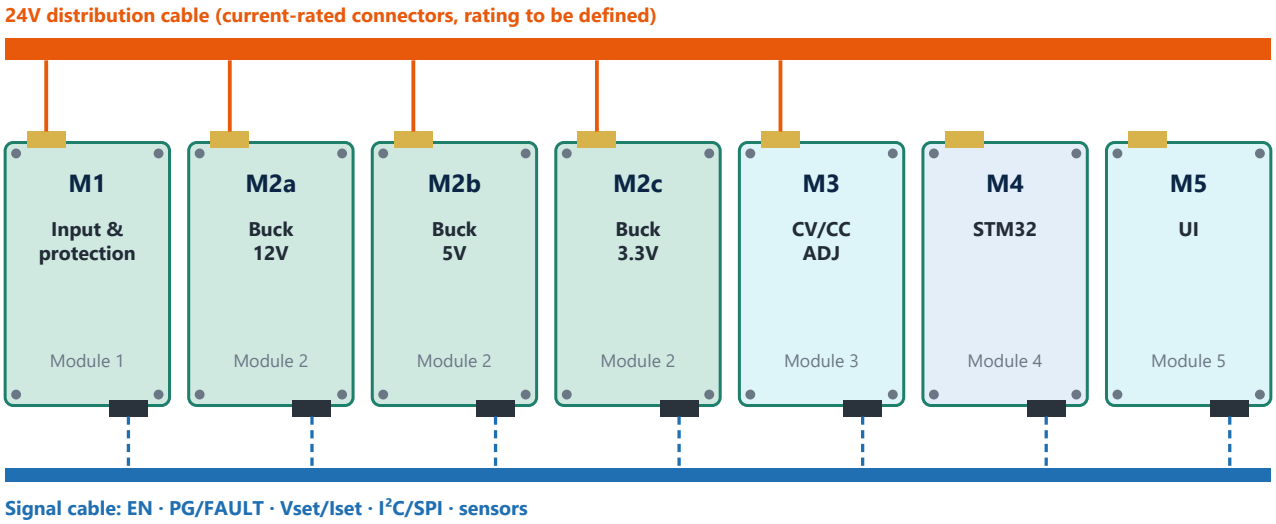
The efficiency values (90%, 88%, 85%, 85%) and the 3W auxiliary power are **preliminary engineering assumptions**. Real efficiency depends on the IC, inductor, switching frequency, load level and temperature; different channels will not reach the same efficiency. Only experimental measurements (section 14) can support conclusions.

PART 07

Hardware Module Breakdown

Hardware Module Breakdown

The system consists of five modules. Each module is one board (or a group of identical boards) connected to the rest through power connectors and signal connectors.



Each module is a separate board, joined with M3 standoffs and detachable connectors, and can be removed and replaced individually.

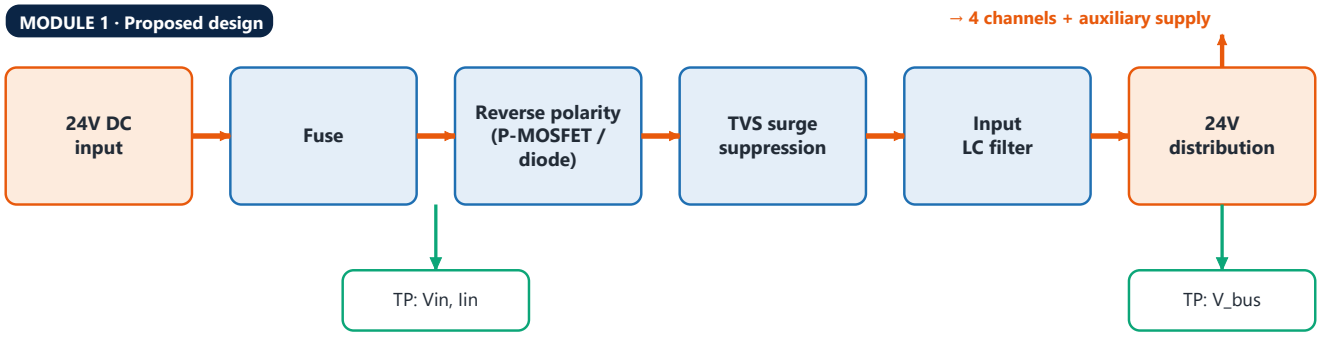
Gold connector = power · black connector = signal. Conceptual illustration, not a detailed design.

Figure 6. Modular board architecture: the 24V distribution cable (orange) and signal cable (blue) connect the modules; each board can be removed and replaced independently. **[Concept model]**

Module 1 – Power Input and Protection

This module receives 24V from the AC-DC supply and "cleans" it before distribution. The main blocks:

- **Fuse:** interrupts current in a serious fault, protecting wiring and the supply.
- **Reverse polarity protection:** prevents damage when the supply is connected the wrong way round; a diode or an ideal-diode P-MOSFET can be used (weighing voltage drop and loss).
- **Transient suppression (TVS):** absorbs short voltage spikes caused by switching or interference.
- **Input filter (LC):** reduces switching noise travelling back to the supply and vice versa.
- **Distribution and measurement points:** splits the 24V to the four channels and the auxiliary supply; provides voltage and current measurement points for the STM32 to monitor.



Multi-layer protection from the input, with voltage/current measurement points for STM32 monitoring.

Figure 7. Functional diagram of Module 1: input protection chain and measurement points. **[Proposed design]**

Module 2 – Fixed DC-DC Converters

Three Buck step-down boards for 12V, 5V and 3.3V, with similar structures to ease design and debugging. The role of each element:

- **Switching IC:** generates the PWM signal and adjusts the pulse width according to feedback; it may integrate the MOSFETs or use external MOSFETs.
- **Inductor:** stores and releases energy, smoothing the current.
- **Capacitors:** the input capacitor supplies pulsed current; the output capacitor holds the voltage and filters ripple.
- **Feedback network:** a resistor divider reports the output voltage to the IC to keep it stable.
- **Switching current loop:** the path of the fast pulsed current through input capacitor – MOSFET – GND; it must be as short as possible (see section 8).

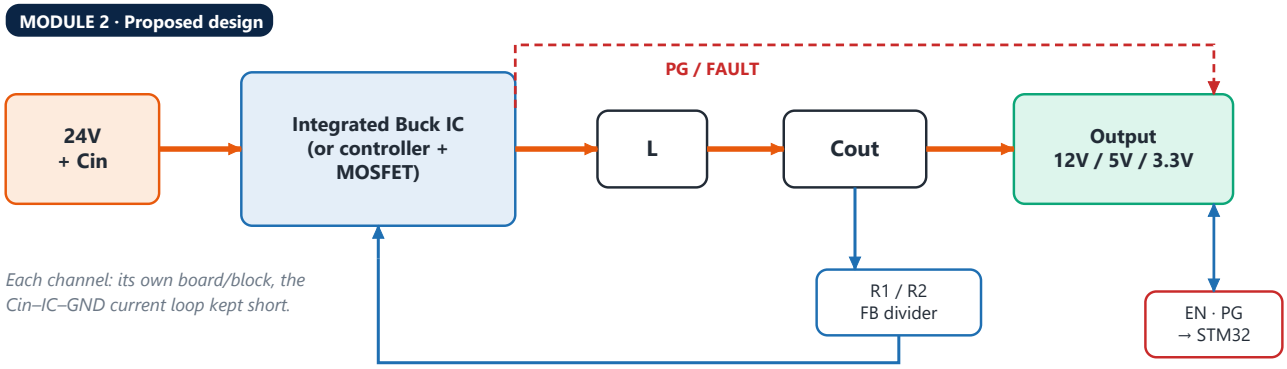


Figure 8. Functional diagram of Module 2: fixed Buck stage, feedback and EN/PG signals. **[Proposed design]**

Module 3 – Programmable CV/CC Power Supply

The fourth channel allows setting the voltage (about 1.8–12V) and the current limit (up to 1A). It operates in two modes:

- **CV mode (Constant Voltage):** the output voltage is held at the set value while the current varies with the load.
- **CC mode (Constant Current):** when the load demands more current than the limit, the circuit switches to holding the current constant and the output voltage falls accordingly.

Example: set 5V / 0.5A. With a 20Ω load the current is only 0.25A, so the circuit is in CV mode (5V). With a 5Ω load, 1A would be needed, exceeding the limit, so the circuit switches to CC: it holds 0.5A and the voltage drops to 2.5V. **This differs from "shut down on overcurrent"**: in CC the supply keeps delivering a controlled current, letting the user test a circuit without resetting each time the limit is reached.

The two control loops (CV and CC) are fast analog circuits; the STM32 only sets the reference values (Vset, Iset) via DAC/PWM and reads back the measurements. An independent analog overvoltage protection is an additional safety layer. The stability of the two loops when changing mode is a topic that **needs study and validation**.

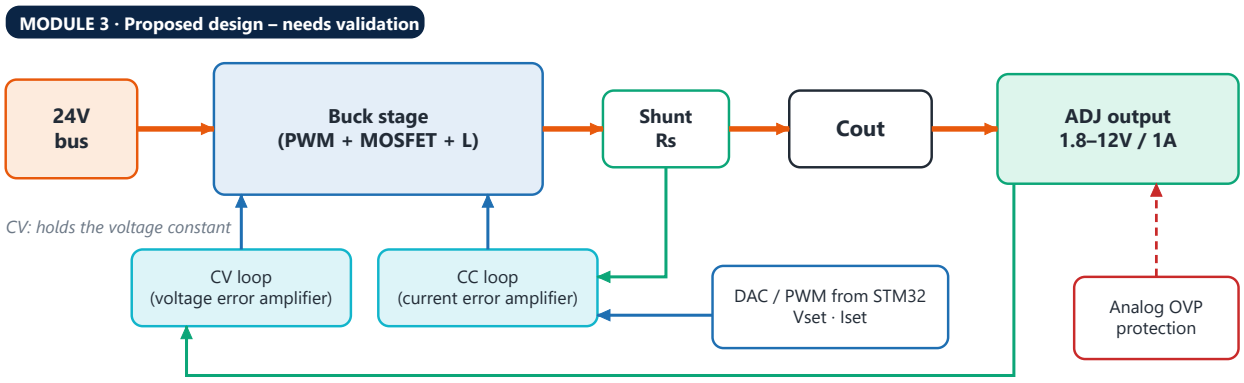
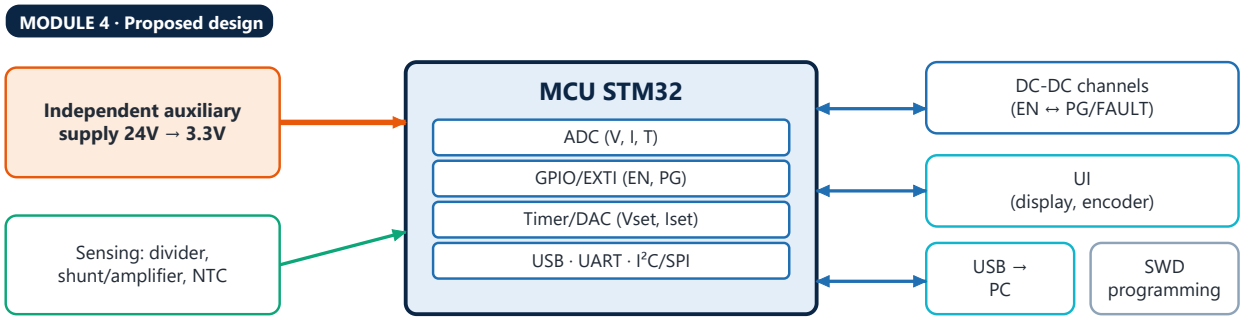


Figure 9. Functional diagram of Module 3: analog CV and CC loops, current-sense shunt, setpoints from the STM32. **[Proposed design – needs validation]**

Module 4 – STM32 Control and Monitoring

- **STM32 microcontroller:** the specific family is not yet chosen; it needs ADC, Timer/DAC, USB and enough GPIO.
- **Measurement circuits:** voltage dividers for voltage; shunt and amplifier for current; temperature sensors (NTC or temperature IC) for temperature.
- **Control GPIO:** EN outputs switch channels on/off; PG/FAULT inputs receive status from the Buck converters.
- **Fault handling:** EXTI interrupt on FAULT, logging, putting the channel into the FAULT state.
- **Independent auxiliary supply, SWD port and USB:** a dedicated supply for the MCU; SWD for programming/debugging; USB for computer communication (RS485 optional).



A separate auxiliary supply keeps the MCU running and reporting when a power channel fails.

Figure 10. Functional diagram of Module 4: MCU, measurement, auxiliary supply and interfaces. **[Proposed design]**

Module 5 – User Interface

The interface consists of a **display** showing the status of each channel (voltage, current, power, ON/OFF/CL mode), a **rotary encoder with push button** for selecting channels and adjusting values, **on/off buttons** for each channel, status **LEDs** and a FAULT LED. The user can also change settings such as ADJ voltage, current limit and warning thresholds. The display type (OLED or TFT) has not been chosen.

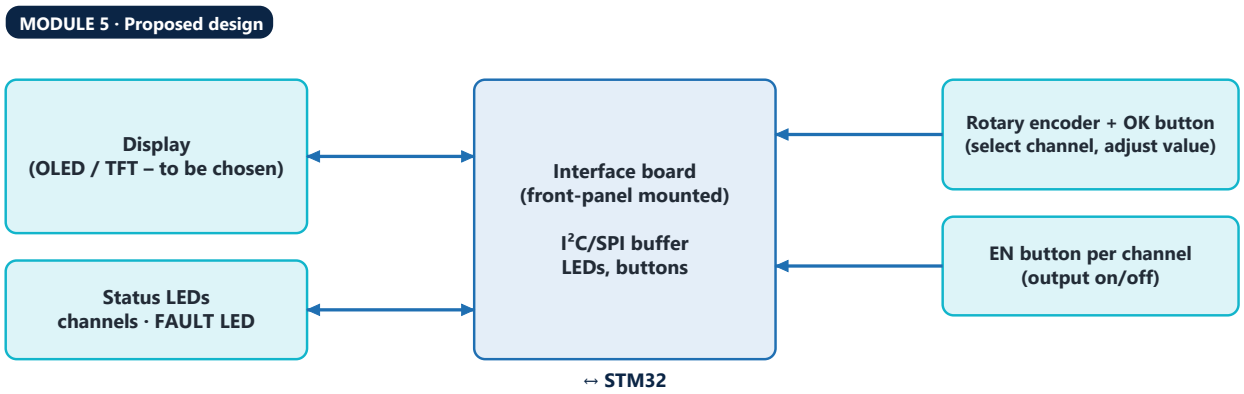


Figure 11. Functional diagram of Module 5: display, encoder, buttons and LEDs connected to the STM32. **[Proposed design]**

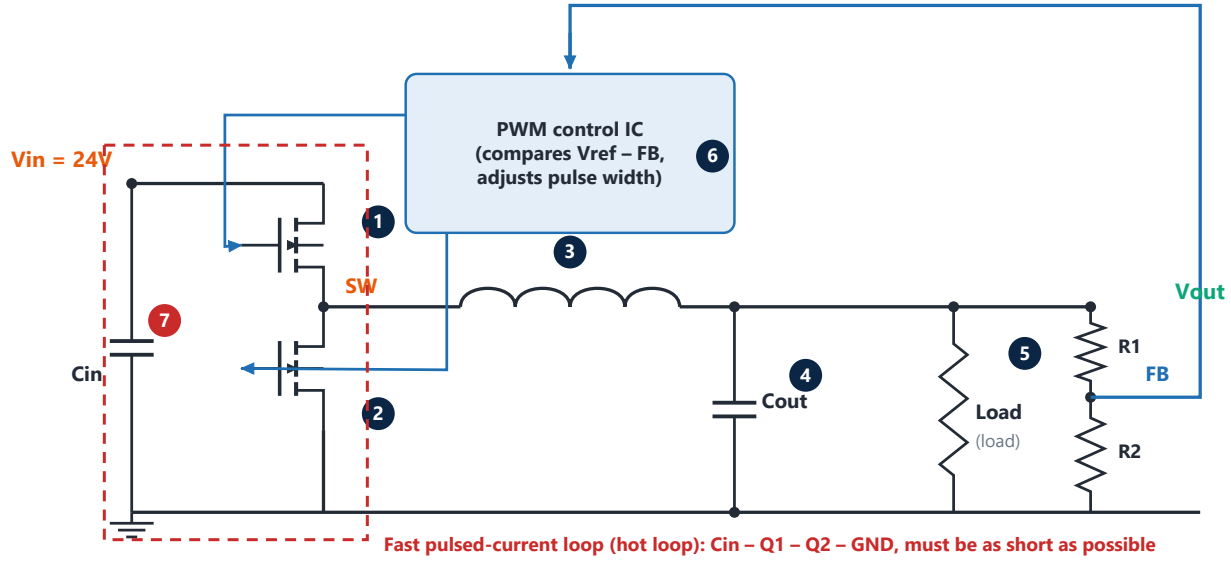
PART 08

How the Switching Power Supply Works

How the Switching Power Supply Works

To step 24V down to 5V, the "easiest" way is a resistor or a linear regulator – but the excess voltage (19V) turns entirely into heat. The Buck converter does it differently: it switches extremely fast (hundreds of thousands of times per second) and uses an inductor to "mix" the energy, so only a small part is lost as heat.

Everyday analogy: imagine pushing a heavy flywheel. You do not need to push continuously – each short push gives the flywheel kinetic energy, and between pushes it keeps spinning. The inductor plays the flywheel of the current: when the switch closes, the current rises gradually and the inductor stores energy; when the switch opens, the inductor keeps "pushing" current through the load.



1. Q1: high-side MOSFET – switches current from Vin
2. Q2: synchronous MOSFET (or diode) – carries inductor current when Q1 is off
3. L: inductor – stores/releases energy, smooths the current
4. Cout: output capacitor – keeps the voltage stable, filters ripple
5. R1/R2: feedback divider – sets Vout
6. Control IC – compares feedback with the reference voltage, modulates PWM
7. Area to optimize in the PCB layout

Figure 12. Simplified schematic of a synchronous Buck converter. The feedback voltage (FB) lets the IC adjust the pulse width to keep Vout stable. **[Principle illustration]**

The two switching states

- **Q1 on:** current flows from the 24V supply through the inductor to the load and output capacitor; the inductor stores energy and the current rises.
- **Q1 off, Q2 on:** the inductor releases its energy through Q2 (or a diode) to the load; the current falls.
- **The output capacitor** smooths the pulsed current into a nearly flat voltage; **the feedback network** measures Vout and adjusts the pulse width to hold the correct value.

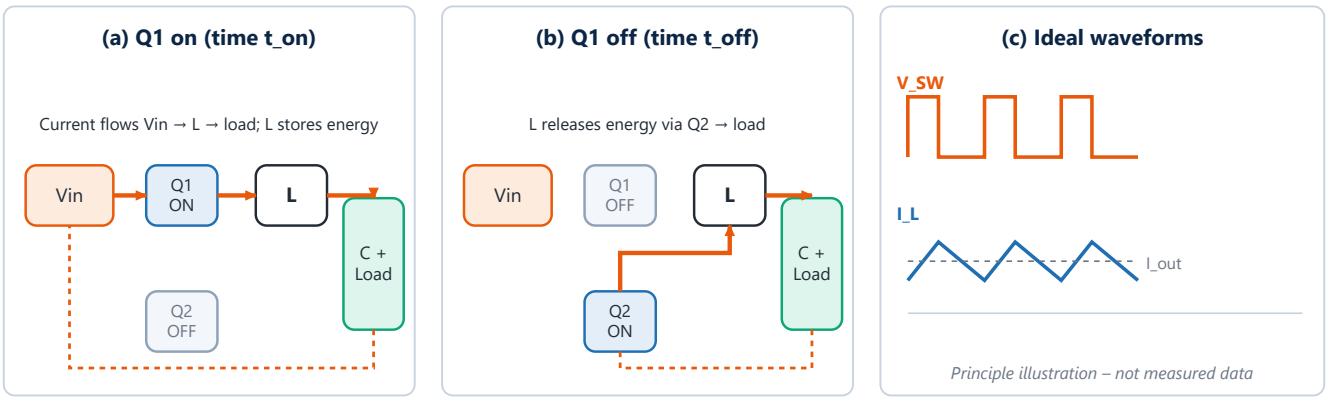


Figure 13. The two states of the Buck converter and the ideal waveforms of the SW-node voltage and inductor current. **[Principle illustration]**

Voltage regulation and losses

In continuous-current mode the output voltage is approximately $V_{out} \approx D \times V_{in}$, where D (duty cycle) is the fraction of time Q1 conducts. From 24V: 12V needs D ≈ 0.50; 5V needs D ≈ 0.21; 3.3V needs D ≈ 0.14. The inductor current ripple is approximately $\Delta I_L = (V_{in} - V_{out}) \cdot D / (L \cdot f_{sw})$. Worked example of the method (not a component selection): 24V→5V, L = 10 µH, f_{sw} = 500 kHz gives $\Delta I_L \approx 0.79$A.

The main losses are: conduction loss in the MOSFET and inductor ($I^2 \cdot R$), switching loss (when the MOSFET changes state), inductor core loss and gate-drive power. Efficiency = P_{out} / P_{in}; each channel has a different efficiency that varies with load.

Why PCB layout makes or breaks the design

In a Buck converter the current changes direction within a few tens of nanoseconds. Every conductor has a small inductance; when the current changes quickly, that inductance causes voltage spikes and noise. Therefore:

- The fast pulsed-current loop (input capacitor – Q1 – Q2 – GND) must be **as small as possible**; place the input capacitor right next to the IC.
- The SW node (large voltage swings) should be kept small in area to reduce radiated noise.
- Sensitive feedback traces must be kept away from the SW node and the inductor.
- Keep the ground (GND) continuous, and choose the connection point between the power section and the analog/MCU section carefully.

> **Practical consequence**
> With the same schematic, two different layouts can give completely different results in noise, heat and stability. That is why PCB layout is one of the focal points of the project, and it can only be judged properly by experimental measurement.

PART 09

STM32 Control Architecture

STM32 Control Architecture

A common question: "Does the STM32 control the whole power supply?" The answer is no. The system is divided into two control levels with different tasks and speeds:

Level 1 – Hardware control and protection: dedicated power-management ICs and analog circuits regulate the PWM themselves, limit current cycle by cycle and shut down on overvoltage or overtemperature. They react quickly and do not depend on software.

Level 2 – STM32 supervision: reads measurements, receives configuration, manages each channel, records faults and communicates with the user/computer. Slower but flexible.

Why not let the firmware do everything?
Software can hang, be interrupted or be delayed by a few milliseconds; in that time a short-circuit current can destroy a MOSFET. The critical protections therefore **must reside in hardware**; the firmware only adds intelligent functions such as fault logging, channel lock-out and notifications.

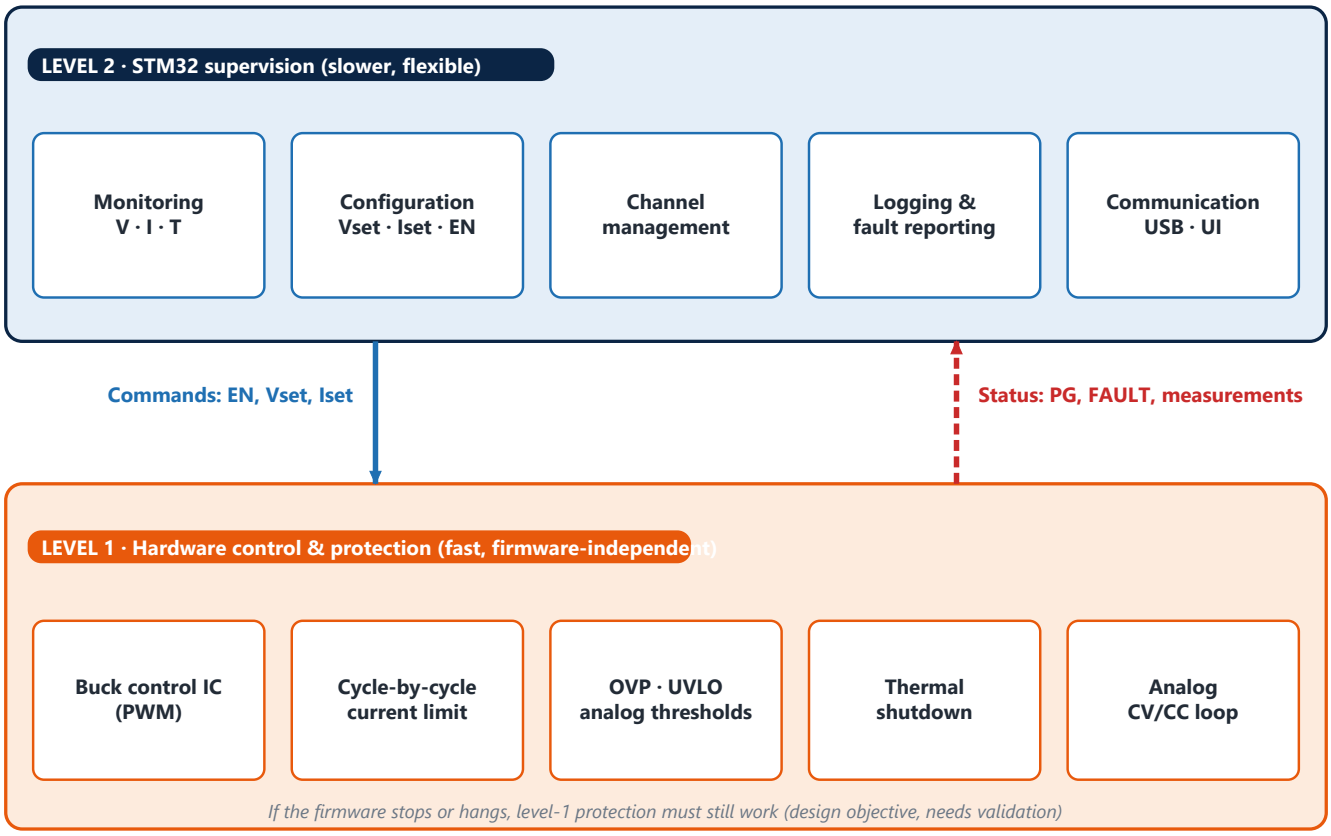


Figure 14. Two-level control architecture. The STM32 sends commands down (blue) and receives status up (red); level-1 protection must work even when the firmware stops. **[Proposed design]**

Simplified firmware architecture

The firmware is organized in layers for easy testing: the hardware layer (HAL) is separated from the service layer (monitoring, logging, configuration storage) and the application layer (state machine, interface, USB protocol). The monitoring period and interrupt priorities will be defined in the firmware development phase.

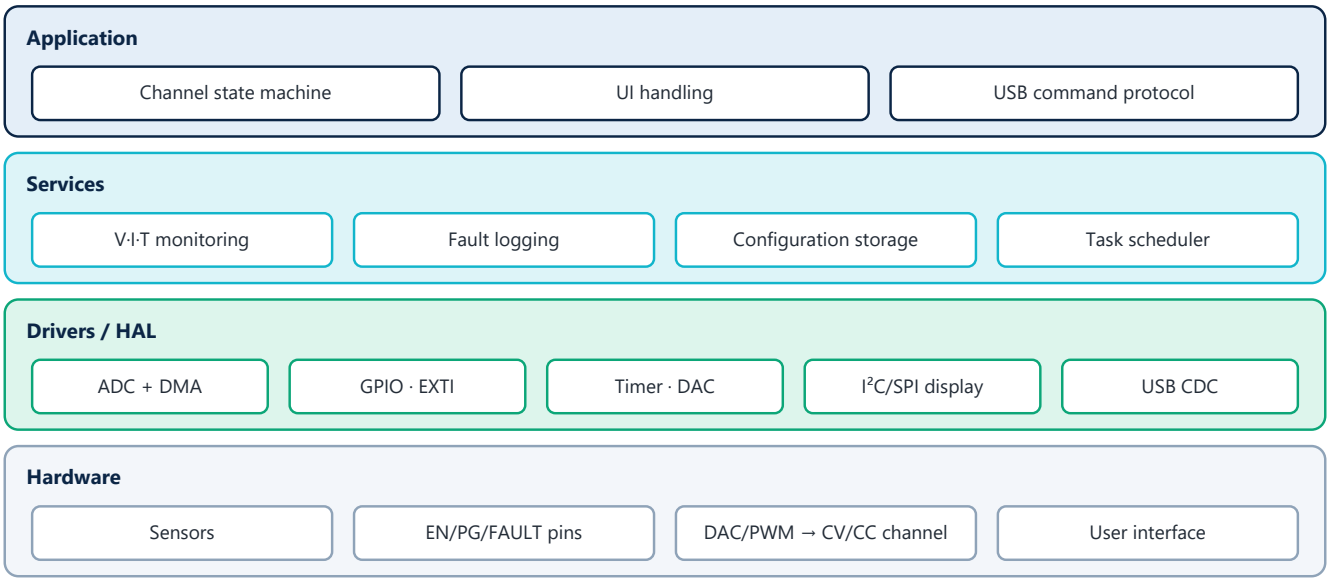
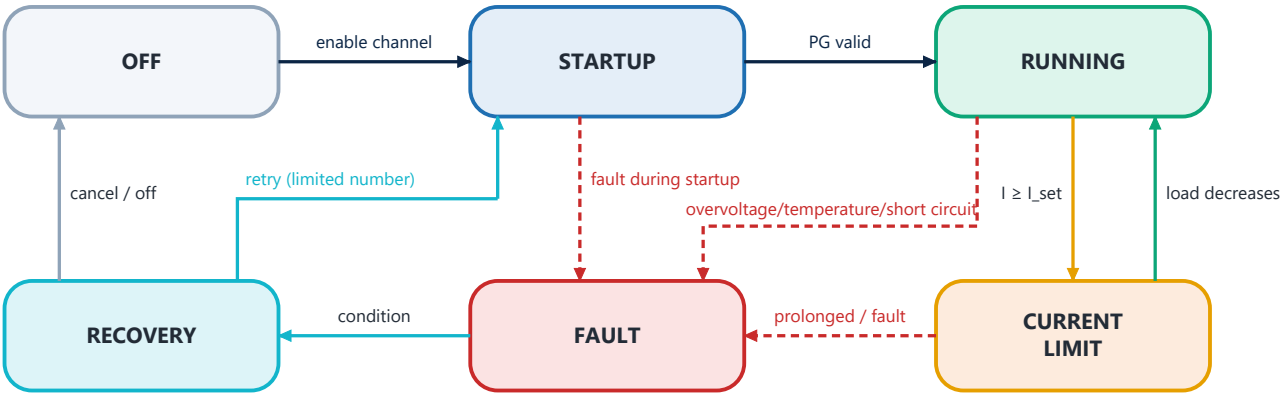


Figure 15. Layered firmware architecture (simplified). **[Proposed design]**

The state machine of each channel

Each channel passes through the states: **OFF**, **STARTUP** (starting, waiting for a valid PG), **RUNNING** (normal operation), **CURRENT LIMIT** (limiting current), **FAULT** (fault, output switched off) and **RECOVERY** (checking conditions for a retry).

Channel state machine – Proposed design



Any state → OFF when the user switches the channel off. Current thresholds, wait times, number of retries and the conditions for leaving FAULT must be defined and validated in later design phases.

Figure 16. Channel state machine: fault transitions are drawn as red dashed lines. **[Proposed design]**

Needs further definition

Current and temperature thresholds, wait times, number of retries and the conditions for leaving FAULT (automatic or user confirmation) are **not yet defined**; they must be specified in the system requirements and validated during testing.

PART 10

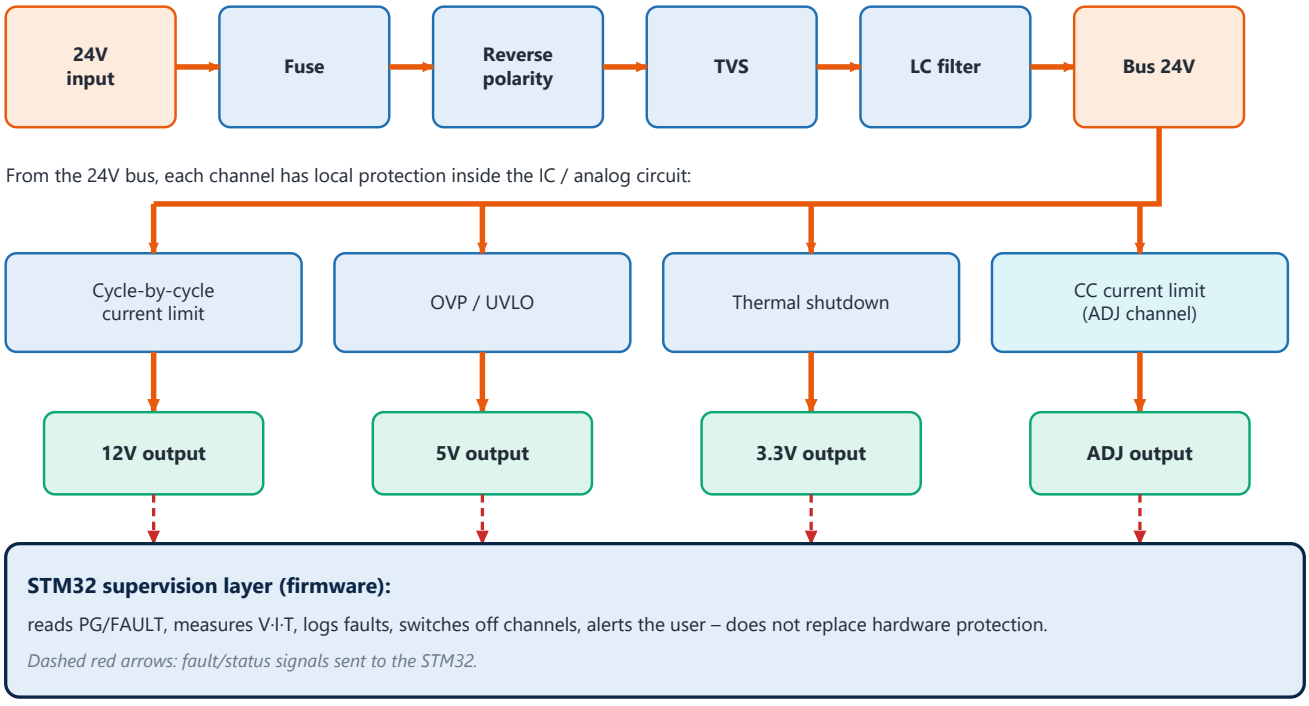
Protection and Safety Philosophy

Protection and Safety Philosophy

Protection is designed in several layers: the input layer protects the whole system, the local layer in each channel protects that channel, and the firmware layer records and reacts at system level.

Protection	Intended mechanism	Responsible
Input reverse polarity	Diode or P-MOSFET reverse polarity protection	Hardware (Module 1)
Input overcurrent	Fuse; input current monitoring	Hardware + STM32 monitoring
Output short circuit	Cycle-by-cycle current limit / pulse skipping (hiccup) in the Buck IC	Hardware (Modules 2, 3)
Output current limiting	Analog CC loop of the ADJ channel (not the same as shutdown)	Hardware (Module 3)
Overtemperature	Thermal shutdown in the IC; board temperature sensor	Hardware + STM32
Overvoltage	Analog OVP threshold; output voltage monitoring	Hardware + STM32
Fault detection and reporting	Read PG/FAULT, log, UI/USB warning	Firmware STM32
Safe startup / shutdown	EN sequencing, wait for valid PG, switch off on fault	Firmware + hardware

Protection chain from input to output (Proposed design)



Thresholds and response times of each layer: Target parameters – Need experimental validation

Figure 17. Multi-layer protection concept from input to output and the STM32 supervision layer. **[Proposed design]**

Fault scenario: short circuit on one channel

Suppose the 5V output is shorted. The expected response occurs in this order: the hardware of the 5V channel limits current immediately; the PG/FAULT pin signals the fault; the STM32 receives the interrupt, logs the event, moves the channel to FAULT and switches off EN; the interface shows a warning. The other channels are **designed to keep operating**.

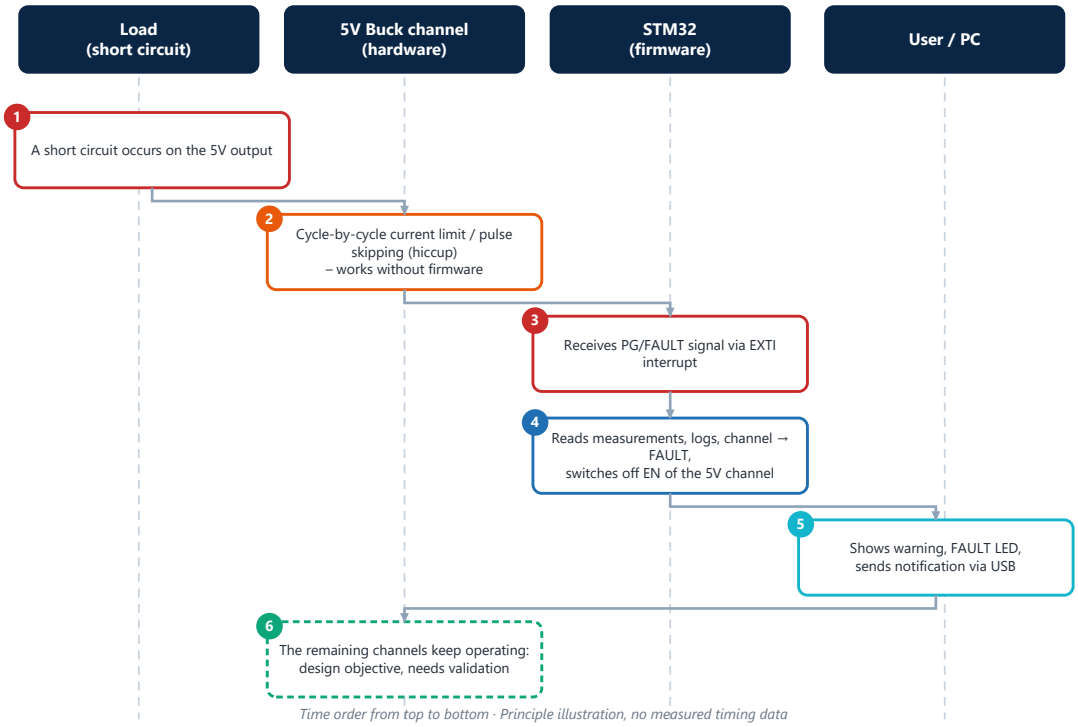


Figure 18. Sequence for handling a short-circuit fault. The fast reaction is by hardware; the firmware records and reports. **[Principle illustration]**

Needs validation

Keeping the unaffected channels in operation is a **design objective that requires validation**, not a demonstrated capability: a 24V bus voltage dip when one channel shorts, noise through the shared GND and the current capability of the AC-DC supply can all affect the remaining channels.

Mains electrical safety

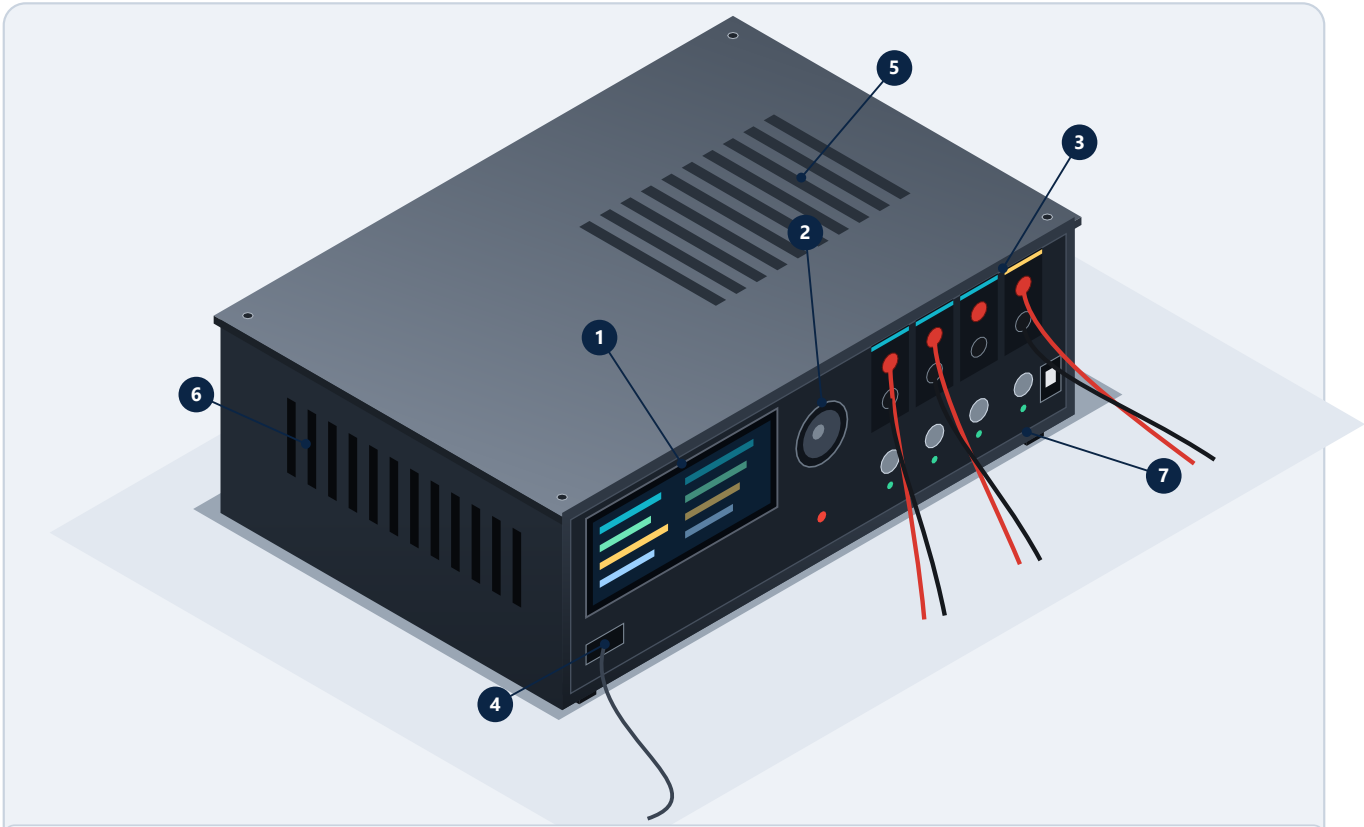
The 230V AC section is handled only by the commercial AC-DC supply and must be enclosed. When mounting it in the case, insulation clearances must be kept, a cable strain relief provided, the metal case earthed where necessary and a suitable fuse fitted. Wiring and testing of the mains part should be done by a knowledgeable person following the applicable safety rules. **This document makes no claim of any safety or electromagnetic-compatibility (CE/EMC) certification.**

PART 11

Modular PCB and Mechanical Construction

Modular PCB and Mechanical Construction

The device is envisioned as a modern desktop laboratory instrument: a compact metal enclosure, a front panel with display and output terminals, and a removable top cover for maintenance – **not** an industrial DIN-rail module. The figures below are conceptual illustrations; board dimensions and mechanical CAD are **not yet finalized**.



1. 4-channel status display
2. Rotary encoder + OK button
3. Four output pairs: 12V · 5V · 3.3V · ADJ
4. USB port to the computer
5. Ventilation slots on the cover (removable)
6. Ventilation slots on the side
7. Non-slip feet

Conceptual illustration – not a photograph of a real device

Figure 19. Impression of the exterior of the device: desktop enclosure, display, encoder, four output terminal pairs, USB port and ventilation slots. **[Concept model]**

Layered assembly structure

The boards are fixed to the base chassis with **M3 standoffs**. The power boards (Buck and CV/CC) sit on the lower level near the AC-DC supply; the STM32 board sits on the upper level close to the front panel to shorten the signal wiring to the interface; the modules are connected by power wires and signal cables with connectors, so each board can be removed during maintenance.

Part	Role	Design note
Removable top cover	Allows maintenance; ventilation above hot components	Fixed with M3 screws (to be defined)
Metal base chassis	Supports the boards and the AC-DC supply; earth reference for the case	Material and thickness not yet chosen
Side wall with slots	Convection air intake	Slots small enough to keep out foreign objects
Front panel	Display, output terminals, encoder, EN buttons, USB	Machined metal or plastic panel

Part	Role	Design note
Rear panel	AC socket, power switch, air exhaust	Cable strain relief; cover for the AC-DC supply
M3 standoffs	Fix the boards and provide insulating clearance to the chassis	Height according to the tallest component
Wires and connectors	Separate power wiring from signal cables	Current ratings to be defined

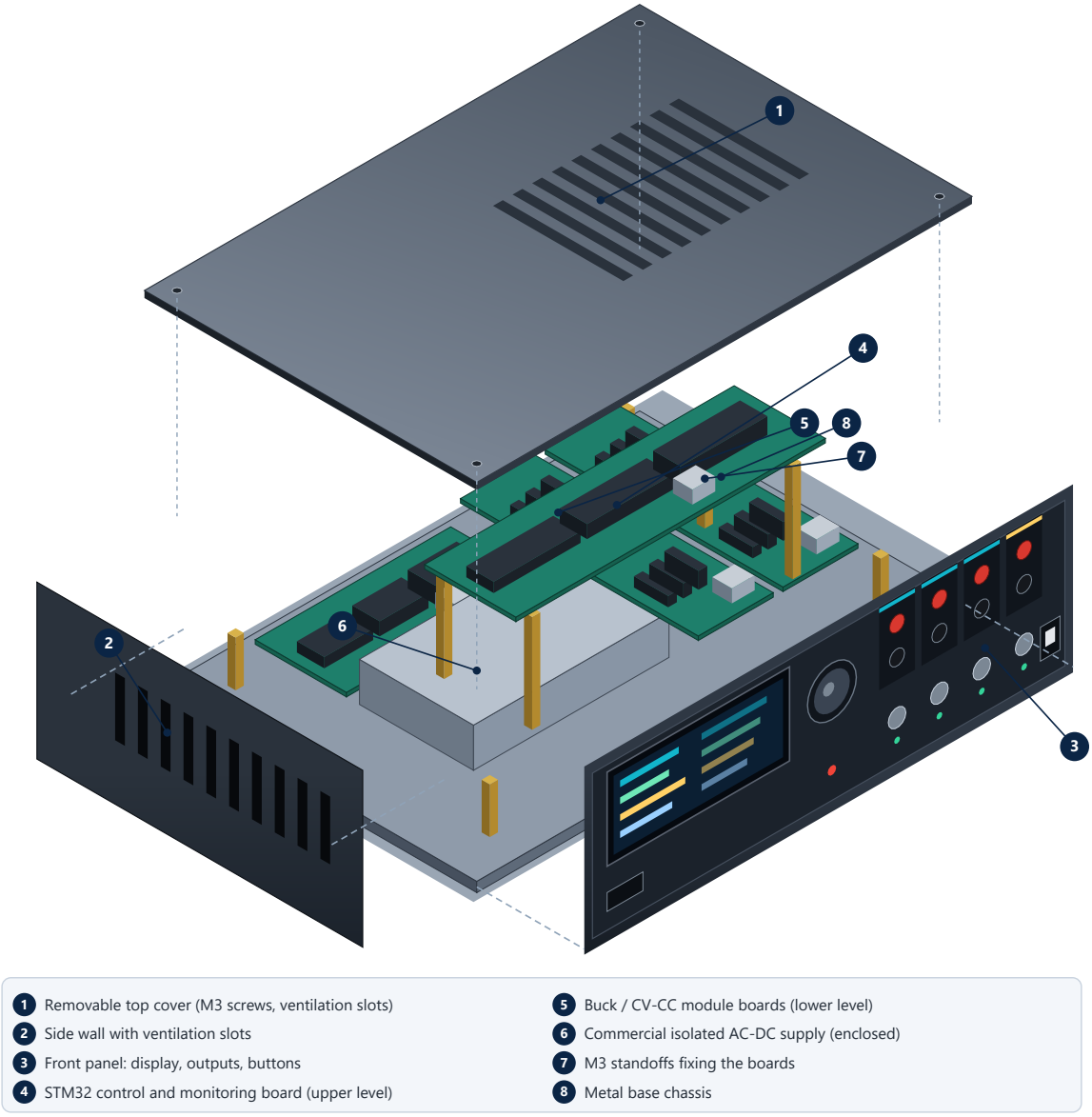
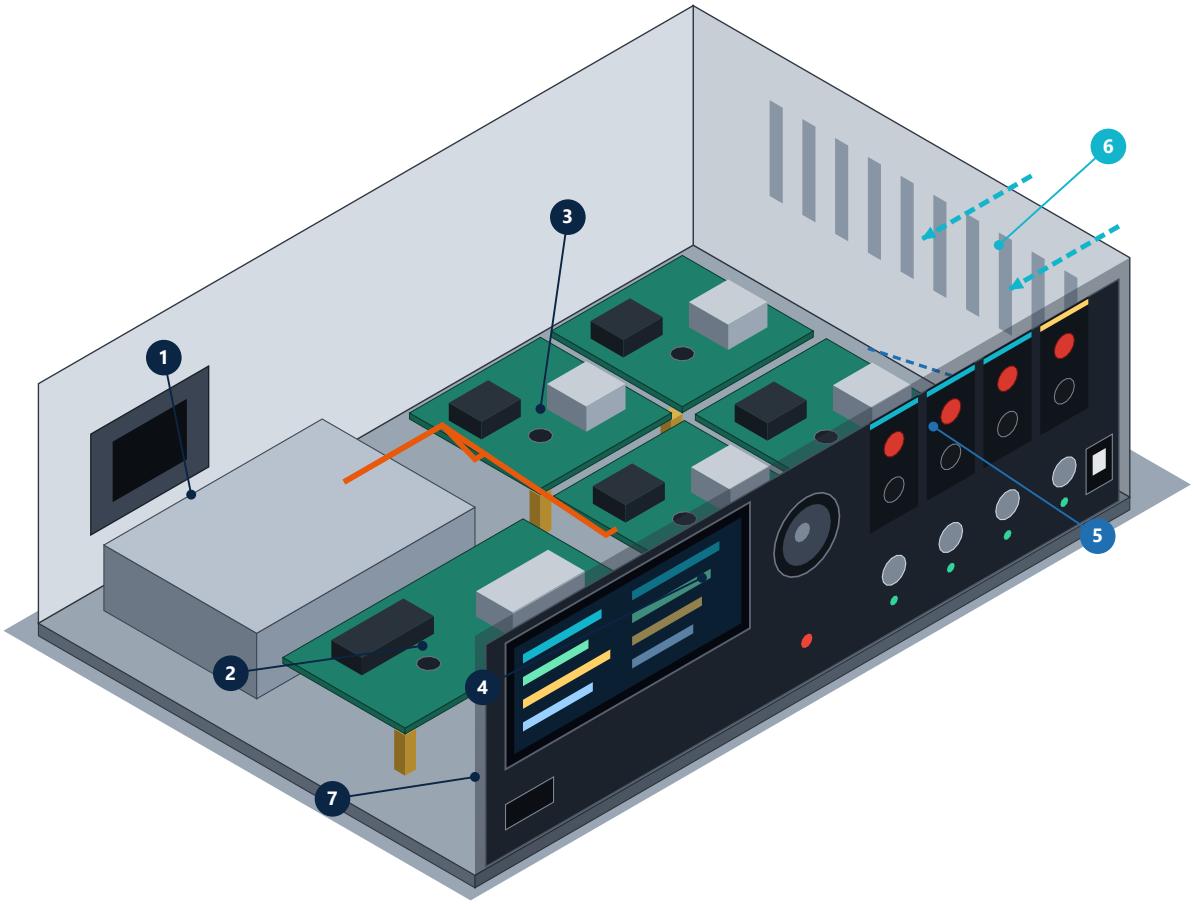


Figure 20. Exploded view: cover, side wall, front panel, board levels, M3 standoffs and base chassis. **[Concept model]**

Internal arrangement

The AC-DC supply is placed at the rear, near the AC socket and covered; the DC-DC boards are arranged in a row so that air flows evenly through them; the STM32 board and the interface board sit near the front panel. The power wiring (orange in the figure) is kept apart from the signal cables (blue) to reduce coupled noise.



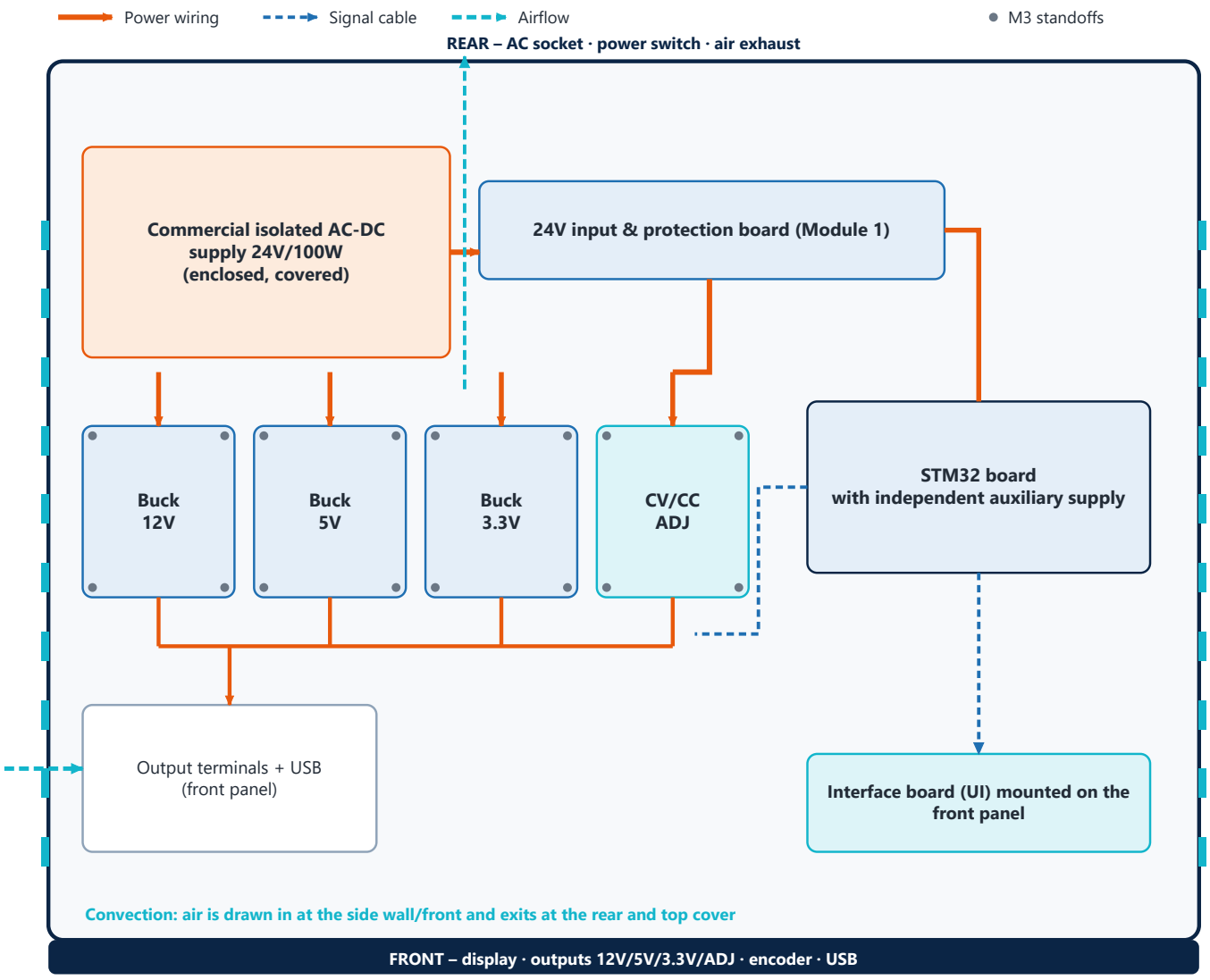
1. Commercial AC-DC supply (enclosed)
2. 24V input & protection board (Module 1)
3. Four DC-DC module boards (Modules 2, 3)
4. STM32 monitoring board (Module 4) close to the front panel
5. Power wiring (orange) · signal cable (blue)
6. Proposed cooling airflow (cyan)
7. M3 standoffs fixing the boards

Conceptual cut-away – board arrangement is not the final design

Figure 21. Isometric cut-away of the interior: AC-DC supply, module boards, STM32 board close to the front panel, wiring and proposed airflow. **[Concept model]**

Mechanical arrangement and cooling diagram

Cooling is by **natural convection**: cool air is drawn in through the side slots, passes over the boards and leaves through the rear and top cover. If thermal measurements show this is not sufficient, a small temperature-controlled fan can be added. The positions of heat-producing components (inductors, MOSFETs, AC-DC supply) must be considered with respect to the airflow.



Illustrative arrangement diagram – not to scale, not final dimensions

Figure 22. Top-view mechanical arrangement diagram with wiring and airflow. Not to scale. **[Concept model]**

Not yet decided

The final board and enclosure dimensions, connector types, enclosure material (aluminum or sheet steel), fan position, and the use of one or two copper layers per board remain open design decisions.

PART 12

Front Panel and User Experience

Front Panel and User Experience

The front panel is laid out so that the user grasps the status of all four channels at a glance: the display on the left, the encoder in the middle and four output columns on the right – each with a label, positive/negative terminals, an on/off button and a status LED.

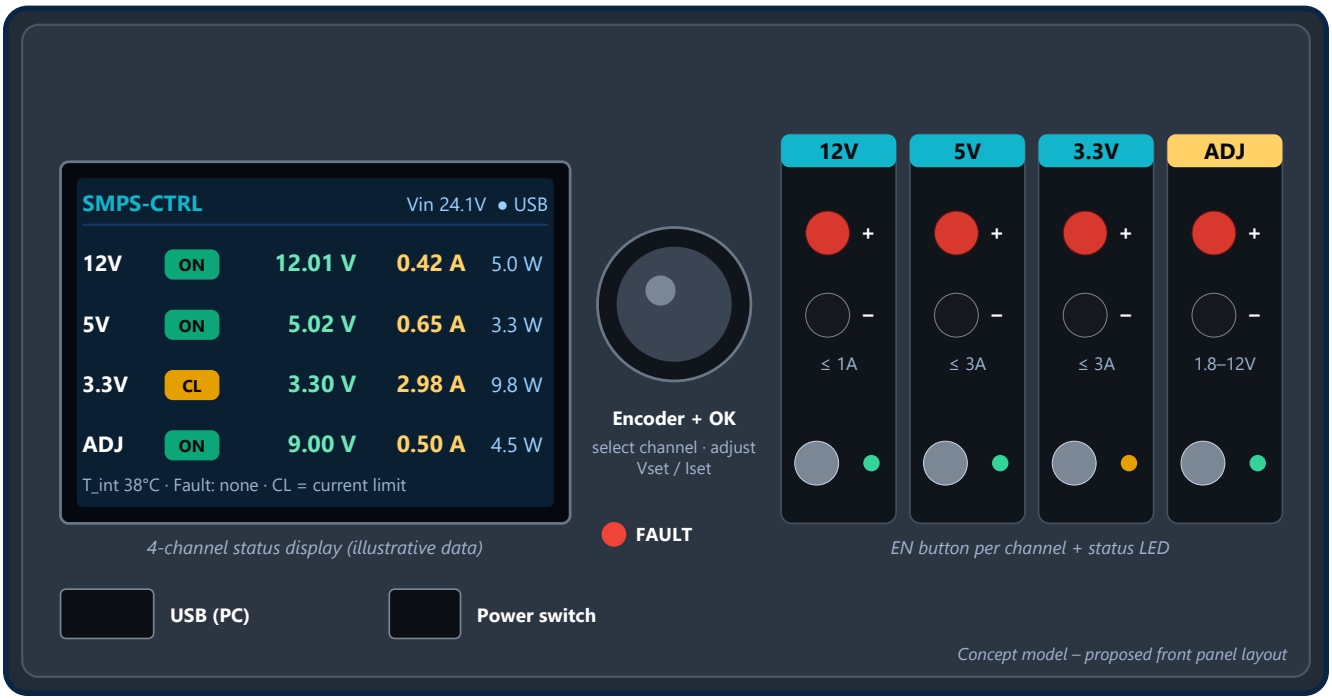


Figure 23. Front panel model (straight-on view): status display, encoder, FAULT, USB, power switch and four output columns. The data on the screen is only illustrative. **[Concept model]**

How the user operates it

- **View status:** the display shows one row per channel – voltage, current, power, mode (ON / OFF / CL / FAULT).
- **Select a channel and adjust values:** turn the encoder to select a channel or parameter; press to confirm; for the ADJ channel adjust Vset and Iset.
- **Switch channels on/off:** a separate EN button per channel; a green LED means the output is live; an amber LED when current limiting; a red FAULT LED on a fault.
- **Connect a computer:** USB port on the front panel for remote monitoring and configuration.

The front panel design aims at practical manufacturability: few unusual parts, standard banana terminals, a commercially available display and common push buttons; the openings can be made in laser/CNC-cut sheet metal or plastic.

PART 13

Development Roadmap

Development Roadmap

The project is divided into six phases, each with clear deliverables and milestones; each phase builds on the results of the previous one. As requested by the project owner, **this document sets no completion dates** for any phase.

1

Phase 1 · Requirements & architecture

Fix system requirements, block diagram, power budget and initial IC selection.

Deliverables: Requirements document (SRS) · block diagram · power budget
Milestone: Architecture reviewed

2

Phase 2 · Fixed DC-DC

Design, simulate, build and measure the 12V, 5V and 3.3V Buck boards.

Deliverables: Schematics · layout · BOM · initial measurement report
Milestone: Three fixed channels running stably

3

Phase 3 · STM32 monitoring board

Control board, independent auxiliary supply, V·I·T measurement, basic monitoring firmware.

Deliverables: STM32 board · firmware v0 · USB communication
Milestone: Correct readings & channel on/off

4

Phase 4 · Programmable CV/CC channel

Design the CV/CC loops, current limiting, analog protection and DAC control.

Deliverables: CV/CC board · test data sheet · calibration procedure
Milestone: Stable CV↔CC transition

5

Phase 5 · Mechanical integration & UI

Enclosure, front panel, display, encoder, module wiring, cooling layout.

Deliverables: CAD drawings · assembly documentation · UI
Milestone: Fully assembled device

6

Phase 6 · Validation & documentation

Carry out the test suite, compile reports, documentation and the source repository.

Deliverables: Measurement report · demo video · GitHub + English README
Milestone: Portfolio documentation complete

Figure 24. Six-phase roadmap from requirements to validation and documentation. **[Plan]**

Phases may partly overlap (for example module PCB design after the schematic has been reviewed), but each milestone counts as reached only when supported by evidence: drawings, measurements or test videos.

PART 14

Testing and Validation

Testing and Validation

A design only has value once it has been measured. Below are the planned tests; each test will have a procedure, equipment and pass/fail criteria defined in advance.

Test	Content	Purpose
Output voltage accuracy	Measure Vout at several load levels	Check the error against the nominal value
Line regulation	Vary Vin within the permitted range	Evaluate stability when the input varies
Load regulation	Vary load current from light to full	Evaluate voltage drop as the load increases
Efficiency versus load	Measure P in and P out at several load levels	Verify the preliminary efficiency assumptions
Output ripple	Oscilloscope, measured at the capacitor pin	Evaluate switching noise
Load transient response	Load step with an electronic load	Evaluate control-loop stability
Startup and shutdown	Observe the EN/PG sequence	Ensure smooth startup without overshoot
Thermal behavior	Thermal camera, temperature sensors	Find hot spots, evaluate cooling
Overcurrent and short circuit	Controlled short of the output	Verify current limiting, shutdown and hiccup
Recovery after protection	Remove the fault, observe recovery	Verify the conditions for leaving FAULT

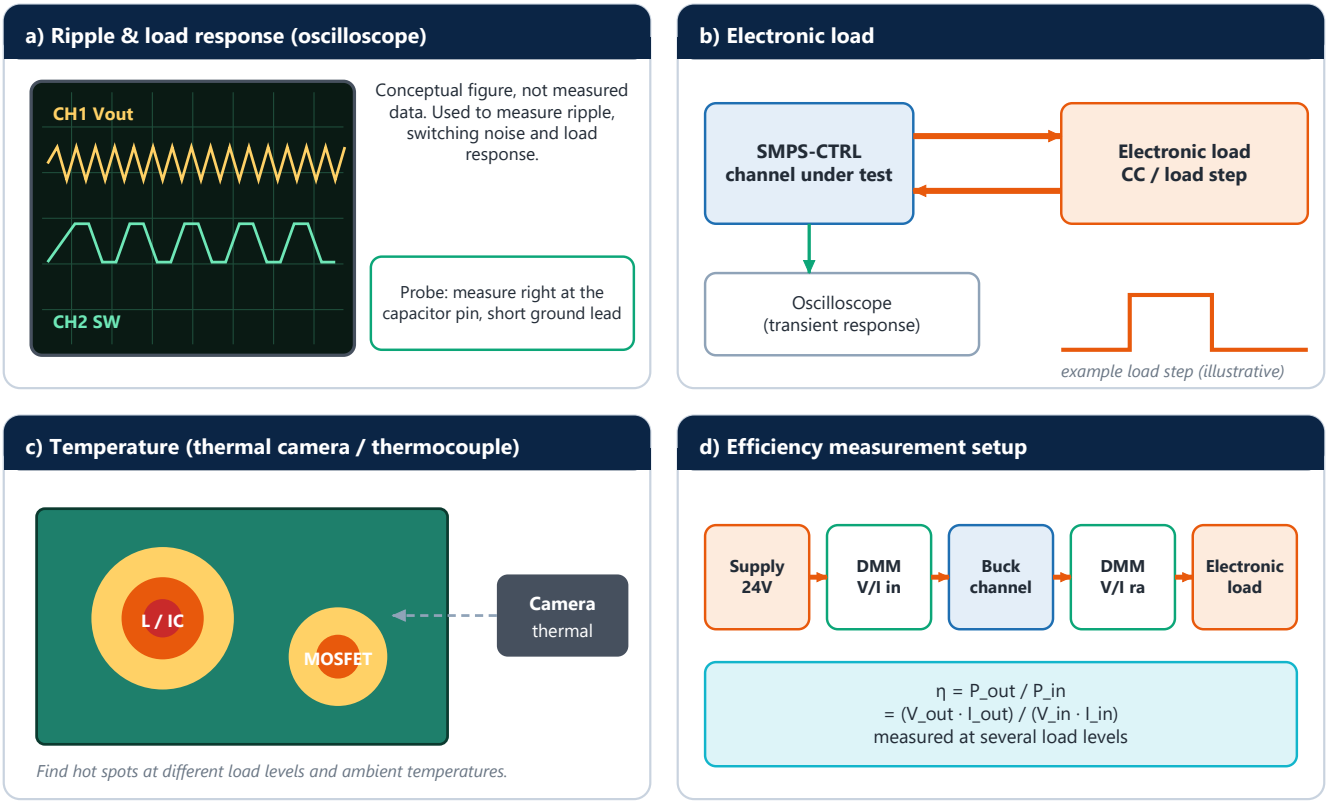


Figure 25. Four panels illustrating the measurement setups: oscilloscope, electronic load, thermal measurement and efficiency measurement. These are not measurement results; the waveforms only illustrate the concept. **[Conceptual illustration – not measured data]**

Principle of honest data
The project commits not to use illustrative graphs as real data. Measurement results will be published only together with the measurement conditions, equipment and uncertainty; any claim about efficiency, ripple or fault isolation will be made only after measurements exist.

PART 15

Engineering Challenges and Design Tradeoffs

Engineering Challenges and Design Tradeoffs

The decisions below are real design problems, each with trade-offs. The "Initial direction" column is a first leaning that may change after testing.

Issue	Main trade-off	Initial direction
Single-layer versus two-layer PCB	1 layer: cheap and easy to make yourself but hard to route, GND less continuous. 2 layers: better layout, less noise, higher cost.	Lean towards 2 layers for the power boards
EMI and switching noise	High frequency reduces the inductor size but increases noise and switching loss.	Moderate frequency; layout and filtering are the focus
Thermal management	Low efficiency causes heat; active cooling adds noise and complexity.	Natural convection, optional fan
Measurement accuracy	The MCU's internal ADC, the reference and the divider error determine accuracy.	Measure first; consider dedicated measurement ICs in a later version
Current sensing	Small shunt: low loss but small signal, noise-prone. Large shunt: good signal but voltage drop and heat.	Shunt + amplifier, sized per channel
Connector current rating	Cheap connectors may overheat at continuous current; this affects reliability.	Choose according to the rating with margin
Modular versus integrated PCB	Modular: easy to repair, flexible but adds connectors and wires. Integrated: compact but harder to improve.	Modules for the prototype
Cost versus performance	Good ICs and quality components raise cost; cheap components raise risk.	Cost acceptable for a portfolio goal
Analog protection versus firmware supervision	Analog: fast and reliable but rigid. Firmware: flexible but slower and can hang.	Two layers: analog as the foundation, firmware as a supplement
CV/CC stability	Switching between CV↔CC easily causes oscillation or overshoot if compensation is poor.	Simulate and test thoroughly in Phase 4

Open decisions

- Buck IC model for each channel, switching frequency and inductor value; the specific STM32 family.
- Commercial AC-DC supply model and how it is enclosed in the case.
- Whether the extended input range of 18–30V is really supported (currently only a proposal).
- Display type, whether RS485 is used, whether a fan is needed.
- Protection thresholds, response times and recovery policy.

PART 16

Expected Final Product

Expected Final Product

When finished, SMPS-CTRL is a desktop device about the size of a small laboratory power supply: a dark metal body, a top cover with ventilation slots and a front panel divided into three zones – display, encoder and four output columns. At the rear are the AC socket and power switch.

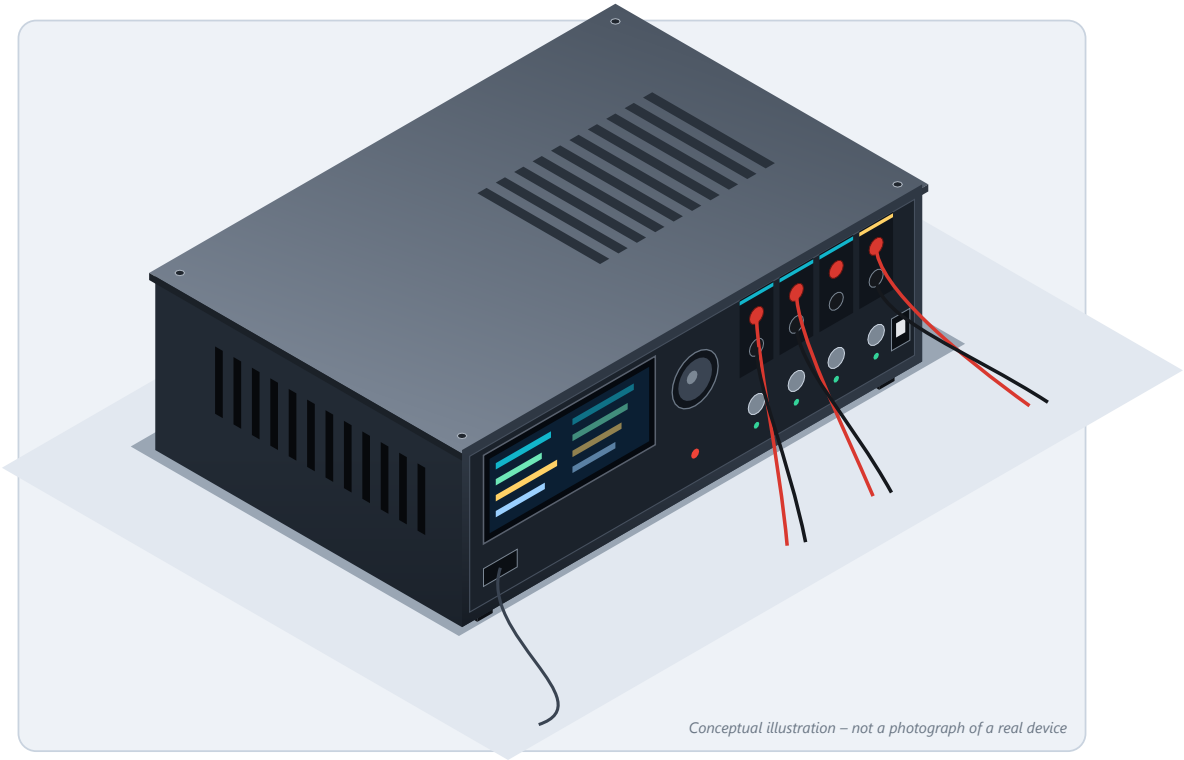


Figure 26. Impression of the finished product on a laboratory bench, with cables powering devices and a USB cable to the computer. **[Concept model]**

How the user interacts with it

Switch on, and the device performs a self-check and shows its status. The user enables each channel with its EN button, connects loads to the terminals and sets the voltage/current limit of the ADJ channel with the encoder. Measurements update continuously; if a fault occurs the FAULT LED lights and the display shows the cause. Connect USB to view measurements on a computer or change the configuration.

How the modules work together

The AC-DC supply delivers 24V → the input protection block cleans and distributes it → four DC-DC converters provide the output voltages → the measurement circuits send data to the STM32 → the STM32 displays it, records faults and receives user commands → commands are passed down to the channels (EN, Vset, Iset). This whole loop runs in parallel with the fast hardware protections.

Applications and value

- Electronics laboratory and teaching: several voltage levels from one monitored device.
- Embedded systems and IoT development: stable 3.3V/5V supply and measurement of the circuit's real current consumption.
- Hardware testing: a safe CV/CC channel for new circuits, reducing the risk of burning components.
- Research and teaching: an open platform for studying power electronics, measurement and monitoring firmware.

Why this is more than "assembling ready-made modules"
The value of SMPS-CTRL lies in what separate modules do not offer: a calculated power architecture (power budget, independent parallel channels), two-layer protection, real-time monitoring, computer communication, in-house high-frequency PCB design, and a fully documented validation process. These are the engineering capabilities needed for hardware, embedded and power-electronics engineering roles.

PART 17

Project Deliverables

Project Deliverables

The following items are the delivery plan; **no item is considered complete at the time of writing this document**.

Group	Deliverable	Main phase
Requirements and architecture	System Requirements Specification (SRS); system block diagram; power budget	1
Electrical design	Schematics; design calculations; simulation results	2–4
PCB design	PCB layout and Gerber files; Bill of Materials (BOM)	2–4
Embedded software	STM32 firmware; USB protocol	3–4
Mechanics	Mechanical CAD drawings; assembly documentation	5
Validation	Test procedures; measurement reports	6
Presentation	Demonstration photos and videos; GitHub repository with English README	6

Each deliverable is version-controlled in the source repository. Measurement reports will state the equipment, conditions and uncertainty; failed results are also recorded as part of the engineering process.

PART 18

Conclusion and Future Development

Conclusion and Future Development

SMPS-CTRL proposes a modular DC power platform for the laboratory: a commercial isolated AC-DC supply providing 24V, independent parallel Buck converters, one CV/CC channel and an independent STM32 supervisor – with fast protection handled by hardware. The technical vision is to turn this idea into a real device, validate it by measurement and present it with a complete set of documentation.

Feasible extensions (not yet implemented)

- Higher-accuracy monitoring (dedicated ADC/measurement ICs, calibration).
- Remote control over USB with a complete command protocol.
- RS485 / Modbus for integration into measurement systems.
- Advanced digital power control.
- Better data logging (CSV export).
- Additional interchangeable power modules.
- Deeper EMI/EMC evaluation.

In summary
This document is a design proposal. It shows the architecture, the reasons for the choices, the technical risks and the validation plan – **not the results of a finished product**.

Appendix – Glossary

Term	Short explanation
AC-DC	Converter from alternating current (mains) to direct current.
Buck	Switching step-down converter: lowers the DC voltage with high efficiency.
CV / CC	Constant Voltage / Constant Current.
EN / PG / FAULT	Enable signal / power-good signal / fault signal.
TVS	Diode that absorbs short overvoltage spikes.
OVP / UVLO	Overvoltage protection / lockout when the voltage is too low.
ADC / DAC	Converter from analog to digital signals / digital to analog.
EXTI	External interrupt of the microcontroller, reacting when a signal pin changes.
SWD	Programming and debugging interface of the STM32.
Hiccup	Protection mode that switches off and retries periodically during a short circuit.
EMI / EMC	Electromagnetic interference / Electromagnetic compatibility.